

## ★ Average

$\Rightarrow$  Equal distribution

10, 15, 20  
 (+5) ← -5  
 15, 15, 15

$$\Sigma n = \frac{n(n+1)}{2}$$

$$\text{Average} = \frac{\text{Sum}}{\text{No. of obsv.}}$$
$$= \frac{n(n+1)}{2}$$
$$\frac{\quad}{nL}$$

$$\star \text{ Avg} = \frac{n+1}{2} = \frac{50+1}{2} = \frac{51}{2} = 25.5$$

Even  $\Rightarrow n+1$   
odd  $\Rightarrow n$  } Avg.

$$\text{Avg} = \frac{\text{Sum}}{\text{No. of Obs (n)}}$$

$$\text{Avg} \times n = \text{Sum}$$

$$\begin{aligned} (n+1) \times n &\Rightarrow n(n+1) \quad \text{Sum of even no.} \\ (n) \times n &\Rightarrow n^2 \quad \text{Sum of odd no.} \end{aligned}$$

Ques: In a school with 15 teachers ~~the~~ the avg. salary monthly is ₹4500. When three teachers left the school, the avg monthly salary decreased by ₹600. Find avg monthly salary of three teachers who left school.

Sol

$n \Rightarrow 15$     avg = 4500     $\Sigma = \text{avg} \times n$   
                                                 $= 67,500$

$n \Rightarrow 12$     avg = 3900     $\Sigma = 46800$

$$67500 - 46800 \rightarrow 20,700$$

$$\therefore \text{avg} = \frac{\text{Sum}}{n} = \frac{20,700}{3} = 6900$$

Note:

Airthmatic Mean  $\Rightarrow$   
Equal distribution among  
given value.

⇒ Average Findings:  
"n", "avg", "sum".

## Small Assignment.

Que. what is average of first five multiple of 12?

(A) 42      (B) 40      (C) 36      (D) 48

Sol 12, 24, 36, 48, 60.

$$\omega_g = 36 \quad \Sigma = 180$$

Que. A <sup>class</sup> day with 20 students has the average age as 14 years. When the teacher is included in the group, the avg. becomes 16 years. What would be the age of Teacher.

Sol  $n = 20$  ; avg  $\Rightarrow 14$

$$n \Rightarrow 21 \quad \text{avg} \Rightarrow 16$$

$$\begin{aligned}\text{Sum} &= \text{Avg} \times \text{no. of obs} \\ &= 14 \times 20 \\ &= 280\end{aligned}$$

$$\begin{aligned}\text{Sum} &= \text{Avg} \times \text{no of obs} \\ &= 16 \times 21 \\ &= 336\end{aligned}$$

Age = 56 years



Que.

The average score of a class of 40 students is 52. What will be average score of rest of the students, if the avg score of 10 of the student is 61.

(A) 47

(B) 49

(C) 50

(D) 48.

(I)

$$n=40 ; \text{avg}=52 \Rightarrow \text{Sum} \Rightarrow \text{avg} \times n$$

$$\Rightarrow 2080$$

$$n=10 ; \text{avg}=61$$

$$\Sigma = 610$$

$$\Sigma \text{ Remaining } 30 \Rightarrow 1,470.$$

$$\text{Avg} = \frac{\text{Sum}}{n} \Rightarrow \frac{1470}{30} \Rightarrow 49.$$

(II)

$$\overset{-3}{A} \quad \overset{-3}{B} \quad \overset{-3}{C} \quad \textcircled{D} \rightarrow \textcircled{52} \rightarrow 52 - 3 \Rightarrow \underline{49}$$

Que.

The average weight of a school of 40 teachers is 80kg. If, however, the weight of the principle be included, the average decreases by 1kg. What is the weight of the principal?

Sol

$$\text{avg}=80 \Rightarrow 3200$$

$$n=40$$

$$3239$$

$$3200$$

$$\textcircled{39}$$

$$\text{avg}=79$$

$$n=41$$

$$\Sigma = 3239$$

Que.

The average age of Abhijeet and Daya is 20 years. Their average age 5 years hence will be.

Sol

$$n=2$$

$$\Sigma = 40$$

$$\text{avg}=20$$

$$n=2$$

$$\Sigma = 50$$

$$\text{avg}=25$$

$$\Rightarrow 25 \text{ years}$$



Que The average monthly salary of 20 employees is 1500. If the manager's salary is added the average becomes ₹1600. The manager's salary is

- (A) ₹3500 (B) ₹3600 (C) ₹3800 (D) ₹3900

Sol  $n=20$  avg = 1500.  $\Sigma = 30000$   
 $n=21$  avg = 1600  $\Sigma = 33600$

$$2000 + 1600 \Rightarrow 3600$$

$$33600 - 30000 \\ \Rightarrow \underline{3600}$$

Que Three years ago, the average of a family of 5 members was 17 years. A baby having been born, the avg of the family is the same today. What is the age of baby?

- (A) 6 months (B) 9 months (C) 1 yr (D) 2 yr. Options

Sol 3 years ago  $\swarrow$   $\searrow$  17+3 = Present  
 $n=5$  avg = 20  $\Sigma = 100$   
 $n=6$  avg = 17  $\Sigma = 102 \Rightarrow 2 \text{ yrs}$

$$(A) y+3$$

$$(B) 2y+9/2$$

$$(C) y+2$$

$$(D) 2y+5/2$$

Que The average of 5 consecutive int. starting with  $x$  is  $y$ . What is the average of 6 consecutive numbers starting with  $(x+2)$ ?

Sol  $x, x+1, x+2, x+3, x+4$  avg =  $y$   
 Note: given integers are consecutive  $\therefore$  all having gap of "1".  
 $\therefore$  middle term is avg  
 $\therefore x+2 = y$   
 Now  $\Rightarrow$  Starting with  $x+2$   
 either we can write "y" instead of " $x+2$ "  
 $\therefore$  6 consecutive.

$$\Rightarrow y, y+1, y+2, y+3, y+4, y+5$$

$\therefore$  all having gap of 1  
 $\frac{n+1}{2}$  even no.

$$\frac{y+2 + y+3}{2}$$

$$\Rightarrow \frac{2y+5}{2}$$